Treball fi de Grau - Curs 2024/2025

Dynamics of rational billiards

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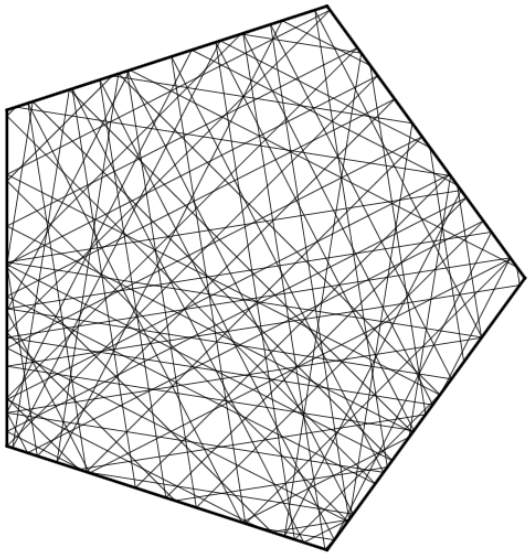
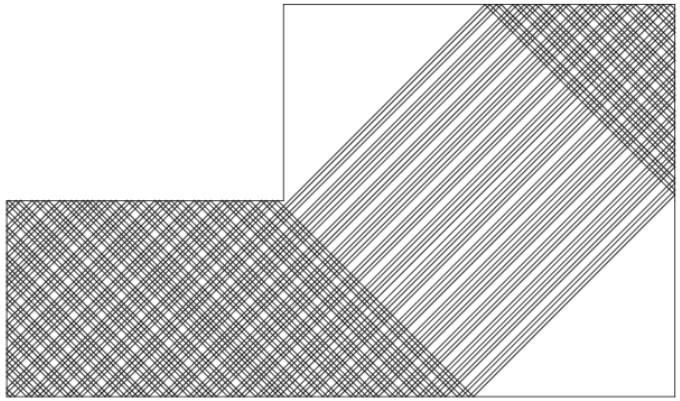
Tutor: Raúl Oset Sinha

5 de setembre, 2025

Facultat de Ciències Matemàtiques

Motivation





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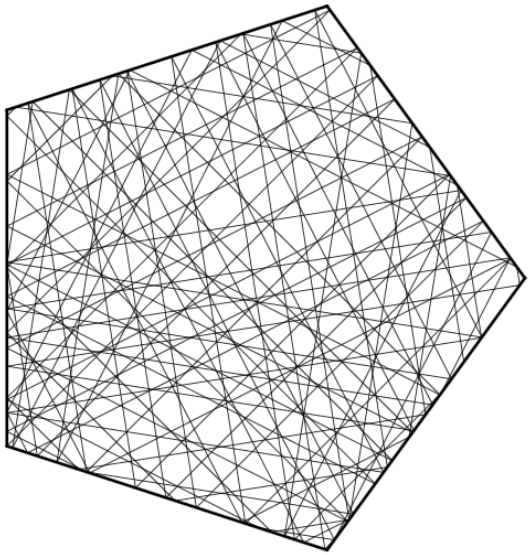

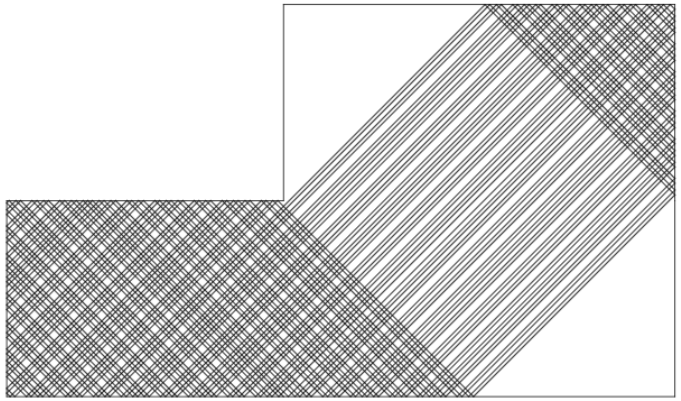

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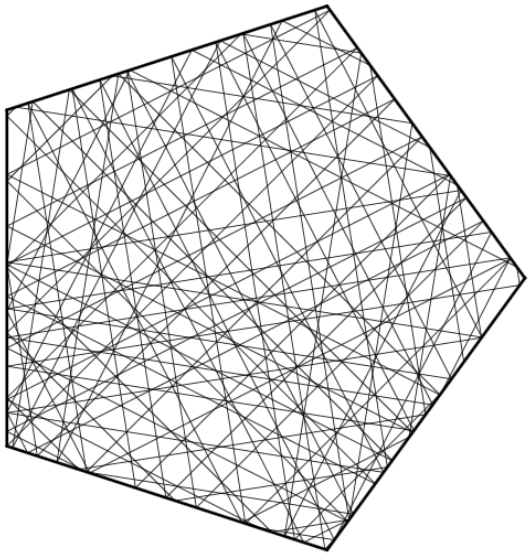

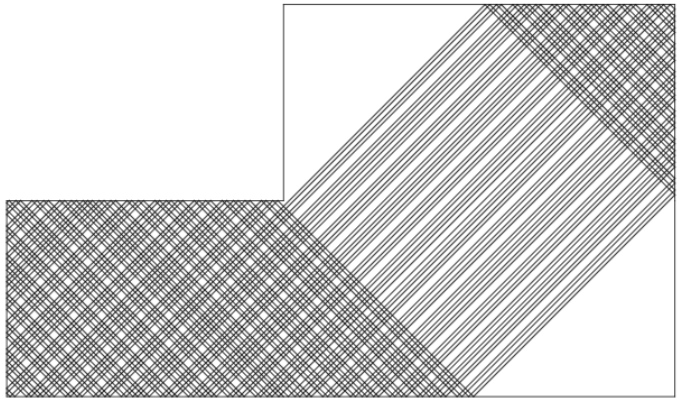

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- Density.
- Periodicity.
- Equidistribution.

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- Each Γ_i is defined by a C^ℓ-map $\gamma_i : [a_i, b_i] \to \mathbb{C}$ ($\ell \geq 3$) parametrized by arclength, which is one-to-one on $[a_i, b_i)$ and has one-sided derivatives up to order ℓ at the points a_i and b_i.

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- Regular boundary points: $\tilde{\Gamma} := \Gamma \setminus \Gamma_*$.

Billiards as dynamical systems

The **phase space** of a billiard system with an underlying table $\mathcal{P}$ is

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We are interested in the dynamical system $(\tilde{\Omega}, \Phi)_{\mathbb{R}}$.

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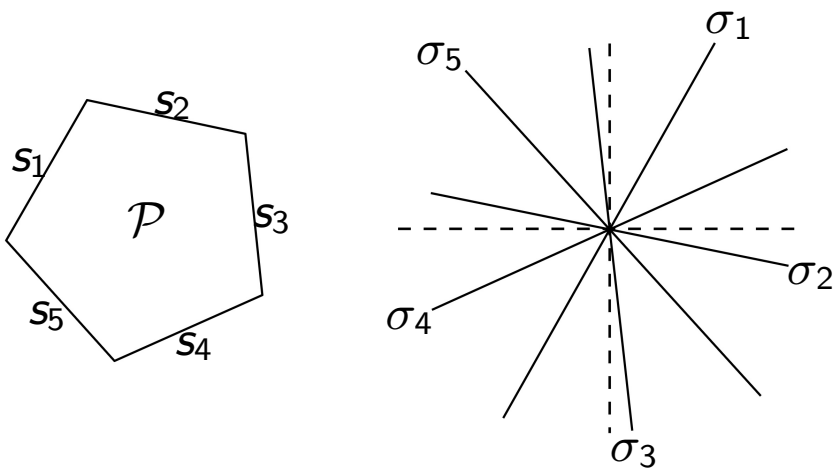

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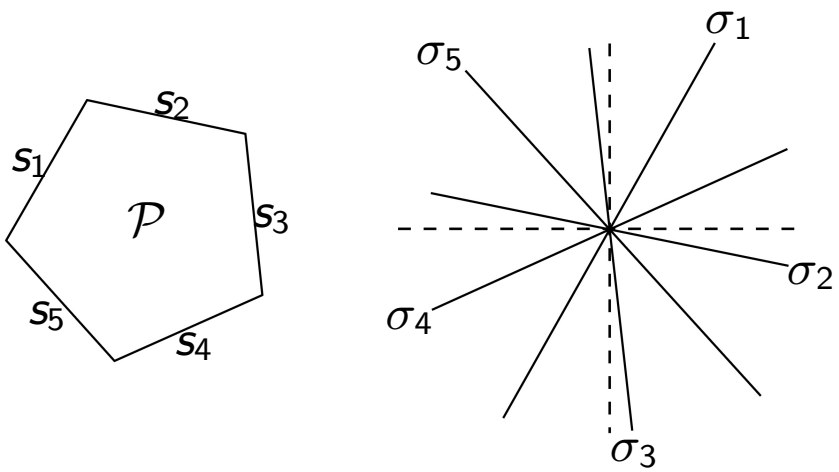

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A polygonal billiard $\mathcal{P}$ is called **rational** if the group G is finite and irrational otherwise.

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3. $R \trianglelefteq G$ and $[G : R] = 2 \Rightarrow G$ is finite.

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- For any two coordinates charts z_α and z_β, we have

$$\phi_\beta(z_\beta)\left(\frac{\mathrm{d}z_\beta}{\mathrm{d}z_\alpha}\right)^2 = \phi_\alpha(z_\alpha).$$

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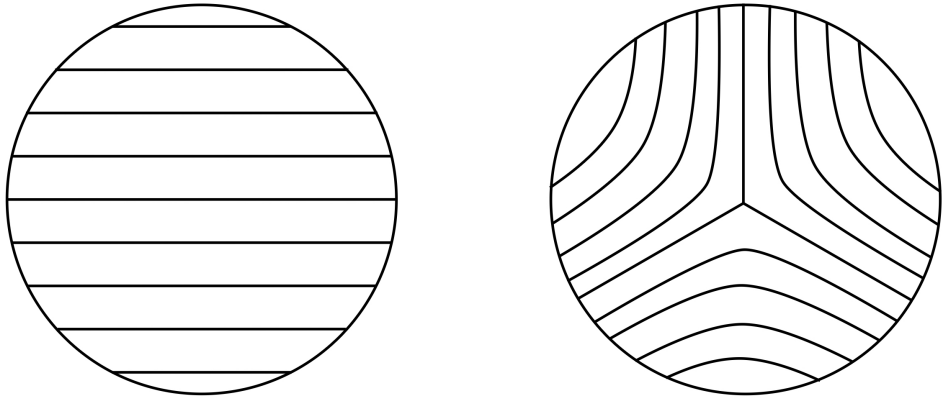

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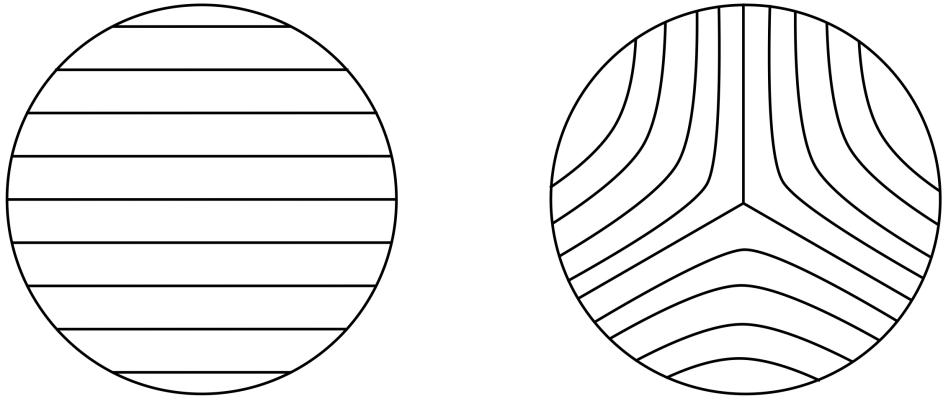

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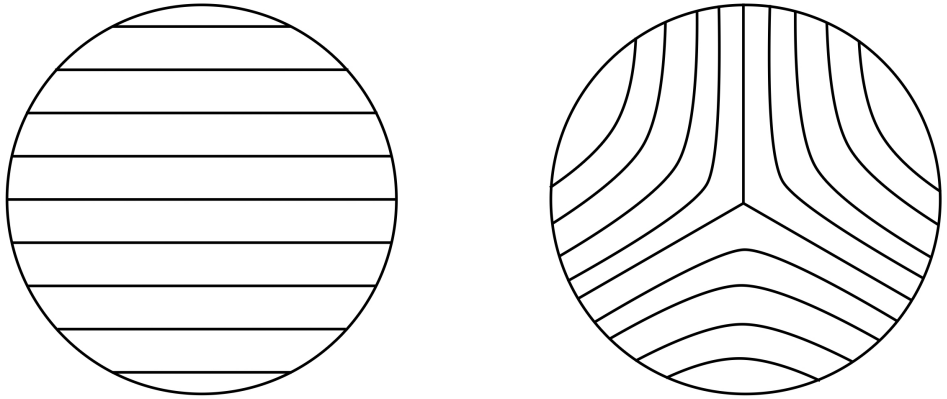

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- The horizontal foliation $\mathrm{Im}\sqrt{q} = 0$.
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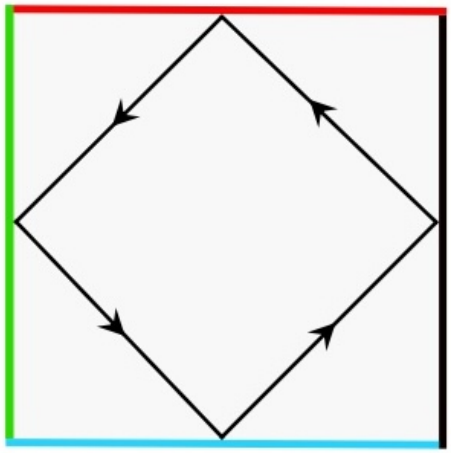

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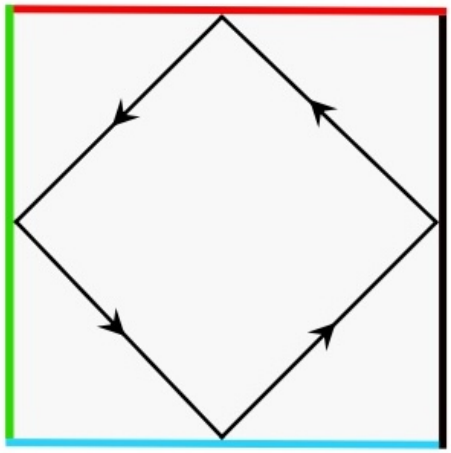

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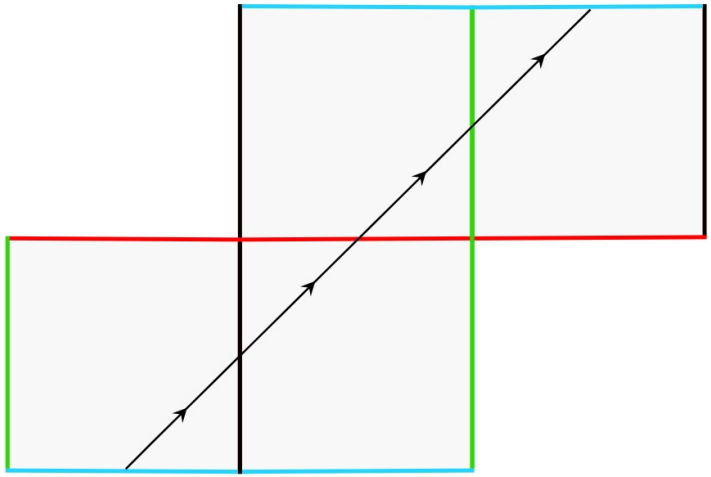

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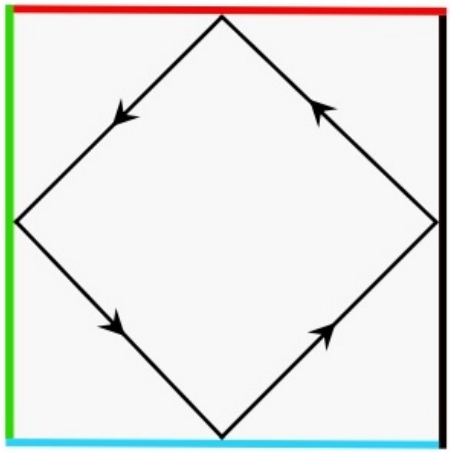

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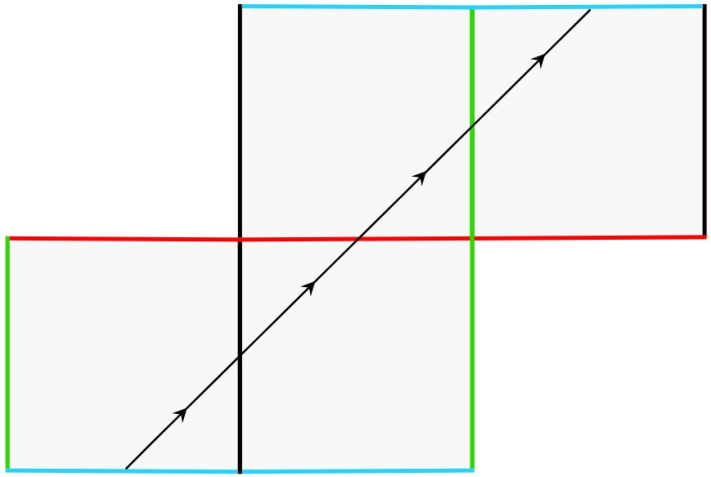

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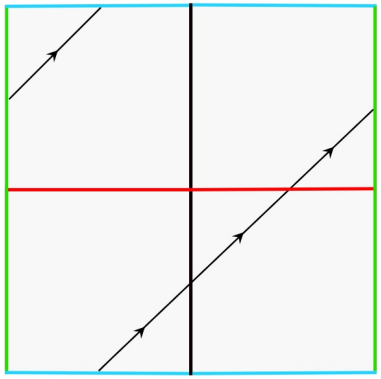

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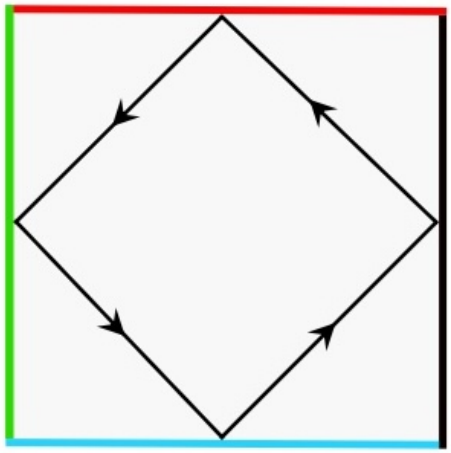

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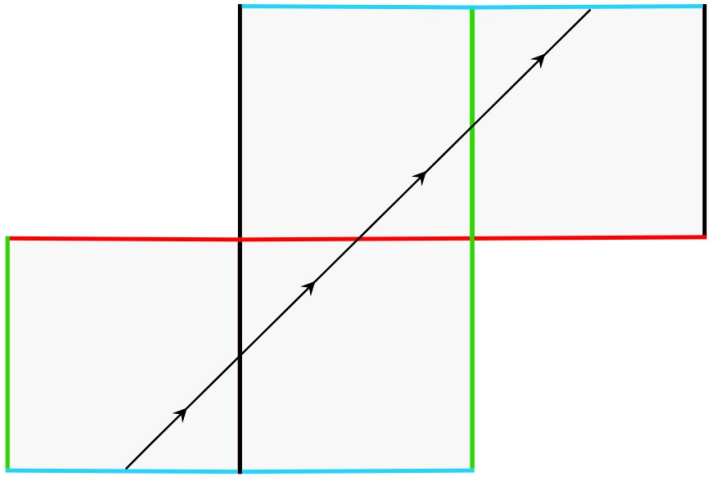

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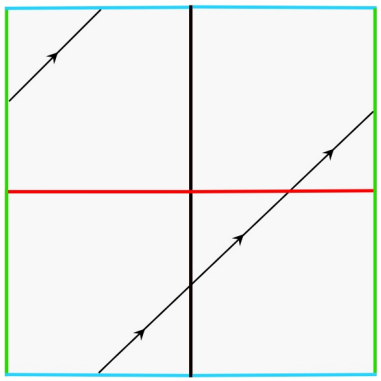

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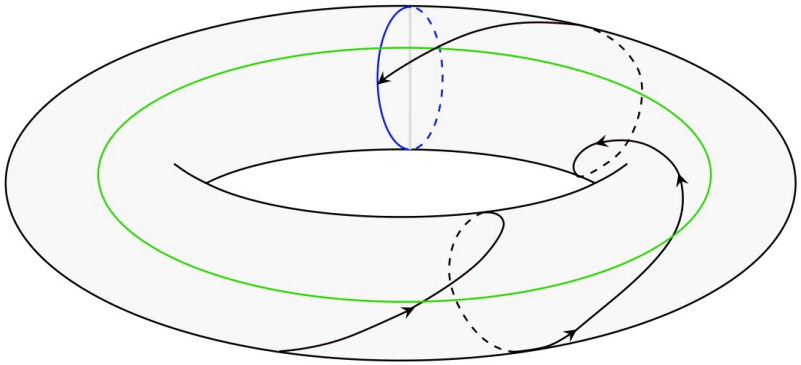

4. Trajectory on the translation surface

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Therefore, proving a statement about quadratic differentials is stronger than proving it for rational billiards.

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- $r_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$.
- $h_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$.

Definition

A quadratic differential q is **divergent** if $g_t(q)$ eventually leaves all compact sets as $t \to \infty$. A quadratic differential is **recurrent** if it is not divergent.

Main theorems (1986, Kerchoff, Masur and Smillie)

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Given a compact Riemann surface M and a holomorphic quadratic differential q, then for almost every θ the vertical foliation of $e^{i\theta}q$ is uniquely ergodic.

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The set of θ's such that $r_\theta(q)$ is divergent has measure zero.

Case I: $r_\theta(q)$ recurrent

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Sketch of the proof:

1. There exists $B \subset \mathcal{Q}_g$ such that if q is not uniquely ergodic then every ω-limit point of the Teichmüller orbit of q is contained in B.

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10. Let $K' = \bigcup_{s\in[-1,1]} h_s(K)$ and $K'' = \bigcup_{k'\in K'} \overline{B(k',1)}$.
11. Let $V_n := \{\theta \in V : g_t r_\theta(q) \in K'' \Rightarrow g_t r_\theta(q) \in B_{\epsilon/2} \quad \text{for } t > n\}$ and call V' one with positive measure.

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16. The density of V' in the interval $[-e^{-t}, e^{-t}]$ is less that $1/2$.
17. Since this holds for a sequence of arbitrarily large values of t, θ' cannot be a point of density for V'. **Contradiction.**

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Q&A

Thanks for your attention